👩🏾‍🏫 Staff Guide · Biology A · Unit 5 Breathing, Exercise & Energy (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in Ms. Dunham's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Unit 3 Diabetes: A2 Exercise Lab, A3 Combustion, A4 Yeast Feast, A5 Combustion & Exercise, A5 Respiration (ATP video + ppt_ATP), Heart/Lungs, A7 Putting It All Together, Metabolic pathways")	**Level 2 piece**
A2 Exercise Lab	Exercise lab packet + notes part A
A3 Combustion + A3.5 Combustion & Exercise (Mallow Out, Liquid on Fire)	Combustion demo packet + notes part B (grouped)
A4 Yeast Feast	Yeast feast packet + notes part C
A5 Respiration (ATP video, yeast balloons) + Heart/Lungs + Metabolic pathways	Notes part D + slides (grouped as one lesson)
Snail/Elodea day 1 + final	Elodea & Snails packet in Unit 2 (her demo sheet was found)
Plant sprout lab	🚧 Under construction (title only)
Class slides	Deck: 19 slides, Words to Know first, hold-up-1-2-3 checks, recap
Her worksheets (grouped into lessons)	Guided notes: 17 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	10 picture cards (cut apart)
—	2 games on the class site: matching + step-by-step order

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no "all of the above"
- Picture anchors: 🔥 cells burn food, 🪙 ATP = energy coins, 🟡 BTB yellow = CO2
- Level 1: no formulas. Level 2: word equation only

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part

4. What to grade

- Quiz stands in for this part of the paper Diabetes test
- **Notes**: same points as her worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver, Room 113 (SLE): changes and IEP questions.
- Facts we could not confirm in her files say "check with Ms. Dunham".

🚧 Under construction (check with Ms. Dunham)

- **Plant sprout lab**: Only a 4-question data quiz; no lab directions.
- **A5 Respiration amounts**: Her yeast-balloon lab leaves the grams of yeast and sugar blank (set by her).

💗 Sensitive topic note

Her A3 Combustion compares a weight-loss contestant (Rachel, Biggest Loser) to a marshmallow and uses before/after photos. We left the person and the photos out: our version talks only about the marshmallow and "our cells." Exercise: follow each student's IEP/504 limits.

6. Photo credits (open licenses)

Human Lungs (NIH BioArt 231 - 630892).png — NIAID / Ryan Kissinger (NIH BioArt), Public domain, Wikimedia Commons · Hand Holdng Fire Toasted Marshmallow (3310040440).jpg — Pink Sherbet Photography from USA, CC BY 2.0, Wikimedia Commons · Running girl at the sports day of Baozhong Junior High School 2015-11-13.jpg — 褒忠國中 雲端網, CC BY 2.0, Wikimedia Commons

👩‍🏫 Staff Guide · Combustion Demo: Mallow Out (Level 2)

Biology A · Unit 5 · goes with the class “A3 Combustion / A3.5 Combustion & Exercise” (5 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Observer card**: A3 Combustion + A3.5 Part 2 “Mallow Out” and Part 3 “Liquid on Fire” (ethanol): teacher demo, our students watch and record. 6 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Smaller or bigger?”
- “Where did it go?”

3. Modifications

- Watch-only role
- Data table given; no molecule modeling (pipe cleaners) required

4. What to grade (5 pts)

- Same 5 points: data + reflection
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

💗 Sensitive topic note

Her A3 asks what “Rachel then/now” (a Biggest Loser contestant) and a marshmallow have in common, with before/after photos. We removed the person and the photos. Talk only about the marshmallow and cells.

6. Photo credits (open licenses)

Hand Holdng Fire Toasted Marshmallow (3310040440).jpg — Pink Sherbet Photography from USA, CC BY 2.0, Wikimedia Commons

👩‍🏫 Staff Guide · Exercise Lab (Level 2)

Biology A · Unit 5 · goes with the class “A2 Exercise Lab” (5 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Lab card**: A2 Exercise Lab: breathing rate, heart rate, and CO2 (BTB) at rest and after 2 minutes of exercise. 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Faster or slower?”
- “Say yellow when you see it.”
- “Why does your heart speed up?”

3. Modifications

- 1 trial instead of 3, no averaging
- Hypothesis with a word bank

4. What to grade (5 pts)

- Same 5 points: data table + reflection
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

💗 Sensitive topic note

Exercise and breathing: check each student's IEP/504 and health plan (asthma, heart, mobility) before the lab. Any movement counts.

6. Photo credits (open licenses)

Running girl at the sports day of Baozhong Junior High School 2015-11-13.jpg — 褒忠國中 雲端網, CC BY 2.0, Wikimedia Commons

👩‍🏫 Staff Guide · Yeast Feast (Level 2)

Biology A · Unit 5 · goes with the class “A4 Yeast Feast” (5 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Lab card**: A3.4 Yeast Feast (2025): 8 mL food + warm yeast in a tube; measure the CO2 gas that collects. 6 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Which tube has the biggest bubble?”
- “Is yeast alive?”

3. Modifications

- Data table given
- Claim with a sentence starter: “The ___ made the most CO2 because ___.”

4. What to grade (5 pts)

- Same 5 points: data + claim + reflection
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

🚧 Under construction (check with Ms. Dunham)

- **This packet**: Her sheet leaves the 4th food as “Your choice.” Please tell us which choice foods are allowed so we can do an allergy check and add the picture.

🫁 Respiration notes (A2, A3, A4, A5)

Biology A · Unit 5 Breathing, Exercise & Energy · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Exercise lab (A2)

1. Exercise makes breathing rate and heart _____ go up.
2. BTB turns from blue to _____ when carbon dioxide is added.
3. After exercise, CO2 production… (circle 1)

goes up goes down

Word bank: **rate** **yellow**

B. Combustion (A3 + A5)

4. Combustion means _____.
5. Combustion releases carbon dioxide and _____.
6. Number the marshmallow lab steps (1-4):

_____ burn it
_____ find the mass again
_____ find the mass
_____ let it cool

Word bank: **burning** **water**

C. Yeast feast (A4)

7. Yeast give off carbon _____ when they use food.
8. Yeast need glucose and _____ for cellular respiration.

Word bank: **dioxide** **oxygen**

D. Respiration and ATP (A5)

9. _____ is energy cells can use right away.

10. The energy in ATP first came from the _____.

11. Match the input or output:

glucose	output (breathe out)
oxygen	output (energy)
carbon dioxide	input (from food)
ATP	input (from lungs)

Word bank: **ATP** **sun**

Level 2 version of the gen ed "A2 Exercise Lab, A3 Combustion, A4 Yeast Feast, A5 Respiration / A3.5 Combustion & Exercise."

📖 Breathing, Exercise & Energy: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Cellular respiration

Cells use glucose + oxygen to make energy (ATP). Carbon dioxide and water come out.



ATP

The energy molecule cells can use right away. Like energy coins.



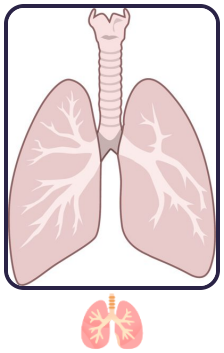
Oxygen

The gas we breathe in. Cells need it to burn glucose.



Carbon dioxide

The gas we breathe out. It is made when cells burn food.



Lungs

Organs that take in oxygen and get rid of carbon dioxide.

Heart

A muscle that pumps blood with oxygen and glucose to the cells.

Combustion

Burning: fuel + oxygen → carbon dioxide + water + energy (heat and light).

Yeast

A one-celled fungus. It eats sugar and makes carbon dioxide gas.

Exercise

Muscle activity. Cells need more energy, so breathing and heart rate go up.

BTB

An indicator. It turns from blue to yellow when carbon dioxide is added.

🔥 Combustion Demo: Mallow Out

Biology A · Unit 5 · Level 2



Name: _____

A3 Combustion + A3.5 Part 2 “Mallow Out” and Part 3 “Liquid on Fire” (ethanol): teacher demo, our students watch and record.

🧺 Demo (teacher does it)

- ☐ 🍡 **1** marshmallow on a skewer
- ☐ ⚖️ **1** scale
- ☐ 🔥 **1** lighter (teacher only)
- ☐ 🕯️ **1** tea-light cup with ethanol (teacher only)

🌱 Steps: check each box when you finish

1. ⚖️ **Record** the marshmallow's mass before burning. ☐
2. 🔥 **Watch** the teacher burn it for a few minutes. ☐
3. ⏳ **Wait** 10 minutes. Record the mass after. ☐
4. ➖ **Find** the difference (before − after). ☐
5. 🕯️ **Watch** the ethanol burn (30 seconds). Record what you see. ☐
6. 💨 **Explain**: the lost mass left as carbon dioxide and water. ☐

📋 Data

Mass before (g)	Mass after (g)	Difference (g)

Ethanol: what did you see? _____

🏃 Exercise Lab

Biology A · Unit 5 · Level 2



Name: _____

A2 Exercise Lab: breathing rate, heart rate, and CO2 (BTB) at rest and after 2 minutes of exercise.

🧺 Lab stuff

- ☐ 🥛 **20 mL** water in a 150 mL beaker
- ☐ 🔵 **8 drops** BTB (staff adds)
- ☐ 🥤 **1** straw (blow OUT only)
- ☐ ⏱️ **1** timer

🌱 Steps: check each box when you finish

1. 🪑 **At rest**: count breaths for 30 seconds, ×2 = breaths per minute. ☐
2. ❤️ **At rest**: count heartbeats for 30 seconds, ×2. ☐
3. 🟡 **At rest**: time how long BTB takes to turn yellow (blow OUT gently). ☐
4. 🏃 **Exercise** 2 minutes (your OK'd move). ☐
5. ❤️ **Right away**: breaths and heartbeats again. ☐
6. 🟡 **Time** the BTB again. ☐
7. 📊 **Compare**: what went up? What went faster? ☐

📋 Data table

Test	At rest	After exercise
Breaths per minute		
Heartbeats per minute		
Seconds for BTB to turn yellow		

One trial is OK (her lab does 3 and averages).

🍞 Yeast Feast

Biology A · Unit 5 · Level 2



Name: _____

A3.4 Yeast Feast (2025): 8 mL food + warm yeast in a tube; measure the CO2 gas that collects.

🧺 Foods to test (her list)

- ☐ 🍬 **8 mL** glucose solution
- ☐ 🍞 **8 mL** starch solution
- ☐ 💧 **8 mL** water (control)
- ☐ ❓ **8 mL** your choice food (staff OK)

🌱 Steps: check each box when you finish

1. 🏷️ **Label** 4 tubes: glucose, starch, water, choice. ☐
2. 🧪 **Pour** 8 mL of each food in its tube. ☐
3. 🍞 **Fill** to the top with warm yeast solution. Cap, thumb over the holes. ☐
4. 🔄 **Flip** each tube into the beaker of warm water. ☐
5. 📏 **Read** the gas every few minutes (to 0.5 mL). ☐
6. 🧠 **Claim**: which food made the most CO2? Why? ☐

📋 Gas (mL of CO2)

Food	5 min	10 min	15 min
Glucose			
Starch			
Water			
Choice: _____			

🚧 **Under construction:** Her sheet leaves the 4th food as “Your choice.” Please tell us which choice foods are allowed so we can do an allergy check and add the picture. *Check with Ms. Dunham.*

🔥 Combustion Demo: Mallow Out · Prep Sheet

Biology A · Unit 5 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
⚖️ **Scale reader**	Reads the mass before and after (after it cools).	
👀 **Watcher**	Says what they see: flame, smoke, color.	
📝 **Recorder**	Writes the masses and the difference.	
📷 **Photo taker**	Before and after photos.	

🧰 Tools and supplies: check when you have it

- ☐ 🥽 **Goggles** everyone
- ☐ ⚖️ **Scale** mass before and after
- ☐ 📋 **Observer card** this page
- ☐ 🧯 **Fire safety** teacher has it ready

⚠️ Safety and manners focus

- 🔥 Only the teacher lights the marshmallow and the ethanol. Stay behind the safety line.
- 🥽 Goggles on. Hair tied back.
- 🍡 Do not eat the marshmallow. (Gelatin and sugar: allergy or diet check if anyone handles it.)
- ⏳ Wait 10 minutes before touching the burned marshmallow.

✅ Before you start

☐ Goggles on ☐ Behind the line ☐ Allergy/diet check done

🏃 Exercise Lab · Prep Sheet

Biology A · Unit 5 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
⏱️ **Timer**	Times 30 seconds for counts and 2 minutes of exercise.	
❤️ **Counter**	Counts breaths or heartbeats (or uses a pulse sensor/watch).	
🟡 **Color checker**	Watches the BTB and says “yellow!” (staff may do the blowing).	
📝 **Recorder**	Writes the numbers in the table.	

🧰 Tools and supplies: check when you have it

- ☐ ⏱️ **Timer** 30 seconds and 2 minutes
- ☐ 🥤 **Straw** one per student, blow out only
- ☐ 🥽 **Goggles** BTB can sting eyes
- ☐ 📋 **Data table** on this card

⚠️ Safety and manners focus

🥤 Blow OUT through the straw. Never suck in. Do not drink BTB.

🥽 Goggles on. BTB can sting eyes.

🏃 Exercise only the way your IEP/504 and staff say is OK. Any move counts: arm circles, chair march, walking.

💧 Drink water after. Tell staff if you feel dizzy.

✅ Before you start

☐ Goggles on ☐ My straw ☐ My exercise is OK'd by staff

🍞 Yeast Feast · Prep Sheet

Biology A · Unit 5 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🧪 **Pourer**	Pours 8 mL of each food into its tube.	
🍞 **Yeast adder**	Fills each tube to the top with warm yeast (staff helps).	
📏 **Reader**	Reads the gas at the pointy end (to 0.5 mL).	
📝 **Recorder**	Writes the gas for each food.	

🧰 Tools and supplies: check when you have it

- ☐ 🧪 **4 tubes** 15 mL cone tubes with caps
- ☐ 🥣 **Beakers** 250 mL + 600 mL, warm water
- ☐ 🌡️ **Thermometer** warm, not hot
- ☐ ⏱️ **Timer** check the gas every few minutes

⚠️ Safety and manners focus

🥽 Goggles on.

🚫 Do not taste the foods or the yeast.

🍬 Allergy check before the lab, especially for the “your choice” food (milk, nuts, gluten, juice).

🧼 Wash hands after.

✅ Before you start

☐ Goggles on ☐ Allergy check done ☐ 4 tubes labeled

🔥 Combustion Demo: Mallow Out · Reflection

Biology A · Unit 5 · Level 2



Name: _____

📸 My photos / proof

☐ Before photo ☐ After photo ☐ Data

✅ Skills I used today (check all you did)

☐ 👀 I stayed behind the line

☐ ⚖️ I found the mass difference

☐ 🤝 I did my role

⭐ Where did the marshmallow's mass go?

1. _____
2. _____

Ideas: carbon dioxide | water | air

🫁 How is burning like what our cells do?

1. _____
2. _____

Ideas: oxygen | energy | CO2 + water

🏃 Exercise Lab · Reflection

Biology A · Unit 5 · Level 2



Name: _____

📸 My photos / proof

☐ Data table ☐ BTB before and after

✅ Skills I used today (check all you did)

☐ ❤️ I measured breaths and heart rate

☐ 🟡 I timed the BTB

☐ 🥤 I used the straw safely

☐ 🤝 I did my role

⭐ What went up after exercise? Why?

1. _____
2. _____

Ideas: **breathing** **heart rate** **carbon dioxide** **cells need energy**

🔥 How is this evidence of cellular respiration?

1. _____
2. _____

Ideas: **CO2 out** **oxygen in** **more energy**

🚧 Plant Sprout Lab

Biology A · Unit 5 · Level 2 · Under construction

Name: _____

🚧 **Under construction:** Only a 4-question data quiz is in the export (hypothesis supported? photo of tubes, plants do photosynthesis and respiration). No lab directions. *Check with Ms. Dunham.*

What we have so far

Quiz Q3 key: animals do respiration; plants do photosynthesis and respiration.

Staff: until this is ready

Have the student do the matching Level 2 unit game or vocabulary cards for this lesson, and give the same points for effort. Mr. Driver, Room 113.

🍞 Yeast Feast · Reflection

Biology A · Unit 5 · Level 2



Name: _____

📸 My photos / proof

☐ Tubes at the end ☐ Data table

✅ Skills I used today (check all you did)

☐ 🫧 I read the gas to 0.5 mL

☐ 🧠 I wrote a claim with evidence

☐ 🤝 I did my role

⭐ Which food made the most CO2? What is your evidence?

1. _____
2. _____

Ideas: **glucose** **mL of gas**

🍞 How does CO2 help bread rise?

1. _____
2. _____

Ideas: **bubbles** **dough gets bigger**

✅ Breathing, Exercise & Energy Quiz

Biology A · Unit 5 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🌿 Animals do respiration. Plants do…

A. Only photosynthesis | B. Photosynthesis and respiration | C. Neither

2. 💨 Yeast need ___ and ___ for cellular respiration.

A. Water, salt | B. Light, carbon dioxide | C. Glucose, oxygen

3. 🫧 The gas collected in the yeast tubes was…

A. Carbon dioxide | B. Oxygen | C. Hydrogen

4. 🍬 Yeast given glucose made more gas than yeast given starch because…

A. Starch is a protein | B. Glucose is cold | C. Glucose is easier to use right away

5. 🪙 ATP is…

A. Energy cells can use right away | B. A kind of fat | C. A gas we breathe

6. 🏭 Most ATP is made in the…

A. Stomach | B. Mitochondria | C. Bones

7. 🏃 Exercise makes heart rate…

A. Go down | B. Go up | C. Stay the same

8. ⚖️ A burning marshmallow loses mass because…

A. Its atoms leave as gas | B. Matter is destroyed | C. It gets wet

9. 🔥 Combustion releases…

A. Oxygen and sugar | B. Only light | C. Carbon dioxide and water

10. 🫁 The lungs take in…

A. Oxygen | B. Glucose | C. ATP

🫁 Respiration notes (A2, A3, A4, A5) · ANSWER KEY

Biology A · Unit 5 Breathing, Exercise & Energy · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Exercise lab (A2)

1. Exercise makes breathing rate and heart **rate** go up.
2. BTB turns from blue to **yellow** when carbon dioxide is added.
3. After exercise, CO2 production… (circle 1)

goes up goes down

Word bank: **rate** **yellow**

B. Combustion (A3 + A5)

4. Combustion means **burning**.
5. Combustion releases carbon dioxide and **water**.
6. Number the marshmallow lab steps (1-4):

2 burn it
4 find the mass again
1 find the mass
3 let it cool

Word bank: **burning** **water**

C. Yeast feast (A4)

7. Yeast give off carbon **dioxide** when they use food.
8. Yeast need glucose and **oxygen** for cellular respiration.

Word bank: **dioxide** **oxygen**

D. Respiration and ATP (A5)

9. _____ **ATP** _____ is energy cells can use right away.

10. The energy in ATP first came from the _____ **sun** _____.

11. Match the input or output:

(Key: glucose → input (from food) · oxygen → input (from lungs) · carbon dioxide → output (breathe out) · ATP → output (energy))

glucose	output (breathe out)
oxygen	output (energy)
carbon dioxide	input (from food)
ATP	input (from lungs)

Word bank: **ATP** **sun**

Level 2 version of the gen ed "A2 Exercise Lab, A3 Combustion, A4 Yeast Feast, A5 Respiration / A3.5 Combustion & Exercise."

🔑 *Key written by our team from Ms. Dunham's materials, please check.*

✅ Breathing, Exercise & Energy Quiz · ANSWER KEY

Biology A · Unit 5 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🌿 Animals do respiration. Plants do… Plant sprout #3

- A. Only photosynthesis
- **B. Photosynthesis and respiration**
- C. Neither

2. 💨 Yeast need ___ and ___ for cellular respiration. A3.10 review

- A. Water, salt
- B. Light, carbon dioxide
- **C. Glucose, oxygen**

3. 🫧 The gas collected in the yeast tubes was… A3.10 review

- **A. Carbon dioxide**
- B. Oxygen
- C. Hydrogen

4. 🍬 Yeast given glucose made more gas than yeast given starch because… A3.10 review

- A. Starch is a protein
- B. Glucose is cold
- **C. Glucose is easier to use right away**

5. 🪙 ATP is… A5 Respiration

- **A. Energy cells can use right away**
- B. A kind of fat
- C. A gas we breathe

6. 🏭 Most ATP is made in the… BL review A6

- A. Stomach
- **B. Mitochondria**
- C. Bones

7. 🏃 Exercise makes heart rate… A2 Exercise Lab

- A. Go down
- **B. Go up**
- C. Stay the same

8. ⚖️ A burning marshmallow loses mass because… A3.5 Combustion

- **A. Its atoms leave as gas**
- B. Matter is destroyed
- C. It gets wet

9. 🔥 Combustion releases… A3.5 Combustion

- A. Oxygen and sugar
- B. Only light
- **C. Carbon dioxide and water**

10. 🫁 The lungs take in… Heart/Lungs

- **A. Oxygen**
- B. Glucose
- C. ATP

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed "A3.10 Diabetes Review, Biggest Loser test review (A2-A6 parts), Plant sprout lab data quiz, A2/A3/A5 sheets". Score: __ / 10.

🔑 *"Plant sprout #3" uses the answer stored in her Canvas quiz. All other answers: key written by our team from her reviews, please check.*